Actuarial digital twins for heatwave failures and thermal ageing in European power insurance

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Research draft | 12 November 2026

Abstract

Warmer summers expose power assets to acute heat stress and accelerate the ageing that makes subsequent failures more likely. We design an actuarial digital twin (ADT) experiment to test whether site measurements improve the estimation and avoidance of insured heat losses. Each asset's operating temperature and condition evolve from its design, duty, age and local weather history. The ADT estimates condition from available measurements using an imperfect thermal model. The class benchmark uses asset class, age, location and historical claims. Within each climate case, we hold the portfolio, policies and weather fixed. We compare estimates of 1-in-200 annual covered loss and a capital proxy at fixed operation, and compare avoided losses at matched intervention spending. ERA5 observations at ten European sites and inverter telemetry establish the pilot exposure and measurement evidence. The paired present and future climate loss comparisons require asset-specific reliability calibration.

Introduction

From 30 January 2027, national measures transposing the revised Solvency II Directive apply. Article 45a calls for long-term climate scenario analysis in the own-risk and solvency assessment of undertakings with material climate exposure, subject to the derogation for small and non-complex undertakings [1]. For power insurance, warming acts through two mechanisms. Heatwaves can cause sudden equipment breakdowns and cluster claims across sites. Higher operating temperatures can also accelerate ageing between extreme events, leaving some assets more vulnerable to later breakdown. A short heat event and years of accumulated thermal damage can contribute to the same insured failure.

We compare a class-based assessment using asset class, age, location and historical claims with an actuarial digital twin using site operating measurements. Claims from earlier years anchor the class baseline. Both approaches receive the climate input for the case being assessed. The ADT estimates each asset's condition by linking measurable operating behaviour to thermal stress, ageing, breakdown, restoration and policy response [9, 7, 8].

The first question is whether the ADT improves the insurer's estimate of the 1-in-200 annual loss and its associated capital proxy. The second is how much loss better operating decisions can avoid. We keep the same portfolio, policy schedules and exposure paths within each climate case. We hold operating actions fixed for the estimation comparison and match intervention spending for the avoidance comparison.

Study design

Asset condition evolves along the temperature history

Local weather, equipment design, load and age drive a physical reference for each asset. The temperature trajectory determines both exposure to short thermal peaks and accumulated damage. The reliability model relates

those states to sudden failures. It follows maintenance, repair and replacement where relevant. The failure event enters the loss account once when high current stress and prior ageing both contribute. Table 1 gives the planned mechanisms.

Table 1. Thermal mechanisms for the asset-specific reliability experiment. The ageing and failure relationships require class-specific operating and reliability evidence.

Asset	Thermal driver	Accumulated condition	Acute outcome
Transformer	Ambient temperature and load drive top-oil and winding hot-spot temperature [2].	Insulation ageing integrates the hot-spot history.	Sudden breakdown risk depends on current stress and insulation condition.
PV inverter	Ambient temperature, power and conversion losses drive cabinet and component temperature.	Capacitor life is consumed at high temperature. Component cycling contributes to solder fatigue.	Breakdown probability rises with consumed life. Reversible protective derating has a separate outcome.
Battery storage	Current and the available cooling capacity govern cell temperature.	Capacity fades and internal resistance rises.	High cell temperature can trigger protection or contribute to damage.
Thermal plant	Cooling-water temperature and flow set the operating margin. River temperature can lag air temperature.	The cooling-water pathway focuses on output limits.	A site-specific discharge or intake limit can force curtailment [4].

The ADT sees measurements available in operation

The class assessment uses class, age, location and pooled claims. The ADT assessment also uses class-appropriate measurements, such as transformer loading and top-oil temperature, inverter power and enclosure temperature, or BMS temperature and current. Its own thermal model estimates the internal state and damage from those measurements. Sensor coverage, model structure and out-of-sample performance determine its error. The reference damage and hidden equipment parameters stay outside both information sets.

Use operating measurements to check the thermal estimates. Fit the insurer's reliance on twin information on a training period of simulated claims, then score its predictions on unseen years or sites. Test the class and ADT assessments on the same simulated outcomes at fixed operation. An asset class may show little or no improvement. Performance on simulated claims describes performance against the reference model. Linked operating and real claims records are needed for field validation of covered losses.

Present and future climate cases use the same insured portfolio

Historical ERA5 drives the present case. Future runs will use bias-evaluated CMIP6 sequences with coherent weather across sites and chronological temperature histories for ageing. Within each climate case, all arms receive the same weather paths and use the same asset register and policy terms. The underwriting comparison changes only the information used to estimate risk under a given operating rule. The intervention comparison changes the operating rule at matched spending and measures the resulting change in damage, curtailment and covered claims.

The policy model applies physical-damage and damage-triggered business-interruption terms to eligible breakdowns. Gradual wear and reversible curtailment affect asset or owner outcomes. A payout for them depends on the policy's actual trigger. Report the annual loss mean, the 99.5th-percentile annual covered loss and its excess over the mean as a study capital proxy. The study will distinguish the present and warming comparisons from the scope of an undertaking's full ORSA. For in-scope undertakings, Article 45a specifies future cases below 2 K and significantly above 2 K of global warming [1]. The existing two-site SSP subset is a data pilot.

Pilot evidence

Heat exposure spans several sites over successive days

Seven of the ten sites reached a daily maximum at least 4 K above their local June–August 90th percentile during 17–21 July 2022 (Fig. 1a). At most four reached that threshold on the same day. Exceedances within that window extend from Almaraz in western Spain to Bełchatów in central Poland (Fig. 1b shows the daily counts). The rate of spells peaking above 4 K increased from 0.37 to 0.72 per site-year between 1991–2005 and 2011–2025 (55 and 108 spells across ten sites in the respective periods, Fig. 1c). This is a descriptive comparison for the selected sites. The daily sequence shows why a definition of the event window matters when modelling claims accumulation.

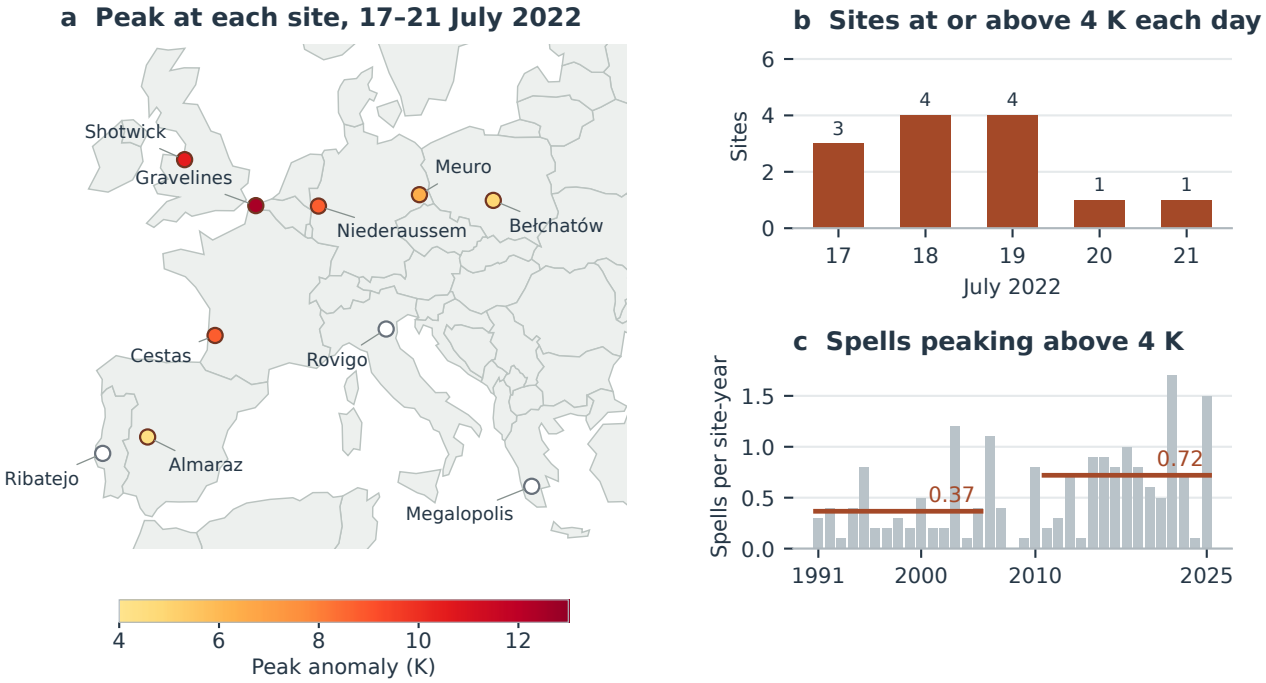


Fig. 1. European heat exposure during 17–21 July 2022. **a**, Peak daily maximum air-temperature anomaly at each of ten sites over the five-day window, relative to its 1991–2020 June–August 90th percentile. Symbols show site values and open markers indicate a peak below 4 K. The maxima may occur on different days. **b**, Number of sites at or above 4 K on each day. **c**, Annual count of site spells peaking above 4 K, divided by ten. Orange segments show the two period means. A site spell lasts at least three consecutive days above its baseline.

The generator reproduces the broad pattern of spell frequencies but its fitted dependence is imperfect. For a site spell peaking at or above 4 K, the fraction of neighbouring peaks reaching at least half that anomaly is 25% in ERA5 and 14% in the model for distances of 800–1,300 km. The observed statistic uses a three-day window around the focal site's peak. The model assigns a shared peak day within each event. Gridded daily histories would allow direct testing of the spatial extent and timing used to aggregate site losses.

Figure 2 shows the paired comparisons that use those exposure histories.

What the pilot establishes

The ERA5 record establishes when and where heat occurs together. The ten locations comprise four solar-labelled plants and six thermal plants. Their daily temperature extrema support exposure analysis. Hourly air temperature and solar radiation are available for all ten sites for June–August 2022. Temperature and damage trajectories also require equipment parameters and operating histories at each asset. Transformer and battery validation need additional asset records, and the thermal-plant mechanism needs local cooling-water observations and operating limits.

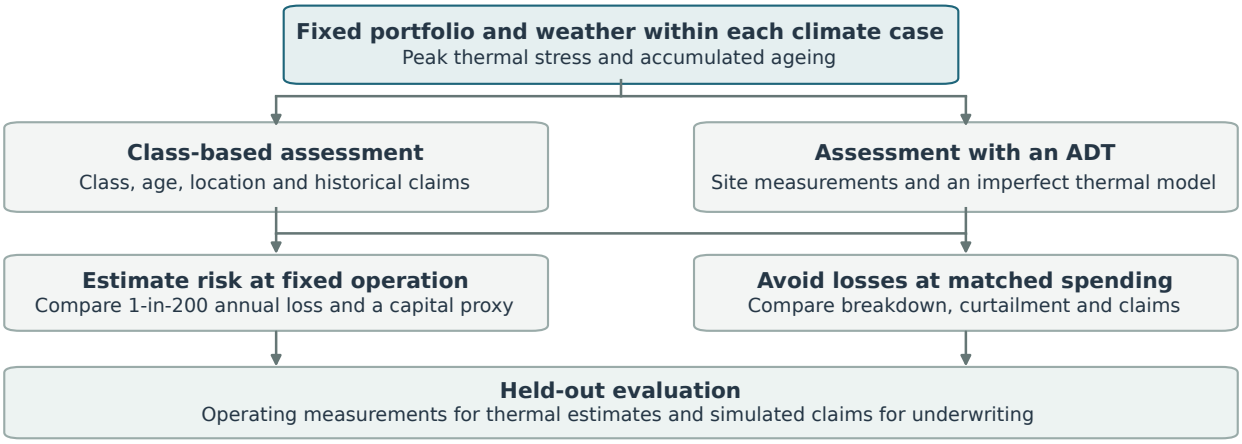


Fig. 2. Paired comparisons within each climate case. The physical reference supplies the loss outcomes. The class and ADT assessments each feed both comparisons. The risk estimation comparison fixes operating actions, and the risk avoidance comparison matches intervention spending.

The inverter field study checks what can be learned from telemetry. In North Macedonia, internal inverter temperature improved five-minute forecasts of self-derating, while the model underestimated coincident starts (Extended Data Fig. 1). The Bali case in DT-INSURE supplies complementary thermal and maintenance evidence [7]. Reversible derating and breakdown are distinct observed outcomes. The current PV code integrates the temperature and capacitor life consumption from a specified history and equipment model [3]. Claims calibration, solder fatigue, asset-level failure probabilities and the paired climate comparison remain to be estimated.

Methods for the pilot evidence

Historical heat and event calibration

We use ERA5 daily maximum 2 m air temperature at ten WRI plant locations from 1991 to 2025. The sites were selected by country and technology strata for a data pilot. The repository records the grid cells, daily UTC aggregation and extraction settings [12, 13]. For each site, we subtract its 1991–2020 June–August 90th percentile and identify runs of at least three consecutive May–September days with a positive anomaly. The July footprint uses the maximum at each site over a five-day window centred on 19 July. Period comparisons count local spells with peaks strictly above 4 K. The dependence diagnostic compares a focal spell peak at or above 4 K with the largest anomaly at each other site within one day of its peak, with ratios clipped to the modelled range.

The synthetic generator uses Poisson event counts, stationary elliptical footprints and a generalised Pareto distribution for the anomaly at each event centre. Simulated moments fit the site-level rates of peak anomalies and distance-binned co-exceedance to the ten-site ERA5 sample. The base fit has about 7.64 events per simulated year and a fitted upper endpoint of 13.1 K at the event centre. The heavier-tail sensitivity uses an endpoint of 15.6 K. Both are properties of the fitted distributions. Event duration and normalised profiles are resampled from observed local spells in three peak-intensity classes and applied across the footprint. Regional event-centre weights remain assumed, and the stationary footprint cannot represent movement over days.

Inverter observations

We use the FusionSolar records in SolarMagix v2 for January–May 2025. Self-derating starts are defined from the vendor state code on the five-minute sampling grid. The weather-and-load and condition models are fitted on January–April and tested on May from the same fleet. Both include plant and unit effects, operating state, recent output and weather. The condition model also uses current internal temperature and a residual from a thermal reference model. Observed fleet counts and expected counts under two dependence structures use the

same eligible windows and unit probabilities [11]. The record covers 37 inverters at six plants and includes 969 bounded self-derating episodes, with a median duration of ten minutes. On the May holdout, adding internal temperature and the thermal residual reduces five-minute onset log loss from 0.01166 to 0.01133. The condition model predicts about 735 starts, with 738 observed. Two plants contribute 736 starts, and outdoor air temperature in the monitored fleet reaches 27 °C. Extended Data Fig. 1 compares coincident starts on this holdout.

Legacy prototype portfolio, breakdown and restoration

Each synthetic book contains 120 utility PV sites, 80 grid transformers, 40 battery storage sites and 25 thermal plants. Each site has a fixed thermal headroom in the simulation, built from a class mean and correlated regional and site terms. The twin estimate adds assumed measurement error. Forecast errors have shared event and site components. A site receives an intervention when its forecast anomaly exceeds the estimated headroom less a trigger margin. The intervention temporarily adds class-specific headroom and incurs a daily cost. For the matched-spending comparison, the class margin is set on the same event sample used for evaluation. Independent evaluation and optimisation of both decision rules remain needed.

Let x_{iek} be positive exceedance above effective headroom at site i, event e and day k. The prototype probability of physical breakdown for that site and event is

$$p_{ie} = 1 - \exp\left[-\lambda_{0,c} \sum_{k} \left(2^{x_{iek}/\theta_c} - 1\right)\right],$$

with assumed class-specific parameters $\lambda_{0,c}$ and θ_c. When the simulation draws a breakdown, it samples damage cost and restoration time from class-specific distributions. Transformer events include a repair and a replacement case. The fitted heat process and these engineering assumptions enter separate stages. All simulated breakdowns are assumed eligible for indemnity. An application requires a failure-mode eligibility check before payout, including the distinction between deterioration as a predictor and a sudden insured breakdown as a trigger [7, 8, 14].

Policy scope and calculation of covered loss

The contract represents the machinery-breakdown portion of an operating power-plant insurance programme. It pays for **physical damage (PD)** from an eligible breakdown and **damage-triggered BI** for the resulting loss of insured earnings during restoration. Operational property policies can also cover damage and consequent BI from fire, storm or flood, depending on the schedule [5, 6]. This experiment's loss totals cover only heat-related equipment breakdown and its consequent BI. Protective derating without damage, thermal-plant curtailment without damage, and intervention spending are treated as owner costs in the prototype. They contribute zero to the covered-loss results.

For a covered breakdown, C is the repair or replacement cost and D is the restoration time in days. With the daily insured interruption amount b, waiting period W, maximum payable days I, combined monetary deductible d and site-event payment limit U, the simulator applies the Actuarial Energy Systems payout convention [9]:

$$B(D) = b \min\{(D - W)^+, I\}, \qquad g(C, D) = \min\{(C + B(D) - d)^+, U\}.$$

Here $x^+ = \max(x, 0)$. The cap I counts payable days *after* the wait. In DT-INSURE the illustrative contract measures the maximum period *from* the date of breakdown and applies a separate PD deductible and limit [7]. Those schedules require a different payout function. For example, under this study's utility PV assumptions, €0.600 million of damage and 45 days of interruption produce 31 payable days. At €15,000 per day, the BI amount before the monetary deductible is €0.465 million. The combined payout is €0.965 million after the €0.100 million deductible, within the €25 million cap.

For an operating portfolio, BI exposure should be estimated from counterfactual plant delivery during restoration and the value insured by the actual wording. This requires plant production and sales records, the failed unit's share of plant output, and deductions for saved expenses or recovery where required. For reversible derating, the

Table 2. Illustrative contract terms used in the saved simulations. The PV daily BI amount is scaled by the assumed plant-size factor. The thermal-plant amount is scaled by capacity in GW. Each maximum counts payable days after the waiting period.

Asset	Daily BI (€/day)	Waiting (days)	Maximum (days)	Deductible (€m)	Limit (€m)
Utility PV	15,000	14	365	0.10	25
Grid transformer	20,000	30	730	0.25	15
Battery storage	25,000	30	365	0.50	60
Thermal plant (per GW for BI)	400,000	30	180	1.00	60

energy shortfall over a period is $\sum_t (P_t^{\text{no derating}} - P_t^{\text{delivered}})^+ \Delta t$, restricted to periods with evidence of derating and using the same realised weather in both outputs. Delivered power can come from telemetry for validation and from a thermal generation model for simulation. Pricing this shortfall at the relevant net electricity value gives the economic loss, separately from breakdown BI. The DT-INSURE weather index uses a generation index H driven by independent weather, a strike K, a payout rate ν and a limit L_W [7]. Its payout is $\min\{L_W, \nu(K - H)^+\}$. Compare the payout with the measured loss to quantify basis risk.

A policy sensitivity should compare PD-only with PD plus BI, vary the eligibility of thermally associated failures, and test the actual waiting period, deductible and limit. The saved runs use the single combined contract in Table 2. Separately estimating the additional loss brought into cover by BI requires applying both contracts to the same simulated damage and restoration events.

Legacy prototype aggregation and underwriting

Each simulated heatwave is an event. The model applies terms per site and event and sums the resulting claims across all sites and events in each year. Average annual covered loss is the annual mean. The 1-in-200 annual loss is the empirical 99.5th percentile of annual aggregate claims. Each book has 20,000 simulated years. The simulator does not track repair carry-over between successive events. An application with overlapping downtime, an hours clause or annual aggregate limits would require chronological simulation and the corresponding contract treatment.

The owner comparison uses common event, forecast, failure and severity draws across operating rules. For the underwriting comparison, both estimates assume the class operating rule. The class model uses the prior distribution of headroom. The twin model updates the joint Gaussian prior with twin estimates and retains spatial covariance. Both predict the same fixed book's class-operation outcome, with per-year draws representing a one-year predictive mixture of uncertain headroom. The annual tail RMSE compares modelled and simulated annual 99.5th percentiles across books. The assumed engineering error and the resulting underwriting comparison provide a model test. They leave open the empirical credibility of the twin on claims.

Data and code availability

The historical inputs and provenance are available in the Heatwave repository [13]. The repository contains the ERA5 daily series, the summer 2022 hourly pilot, the field-study summary, the legacy simulated books and the figure script. It also contains a PV thermal and capacitor-ageing kernel for the asset-specific experiment. SolarMagix v2 data are available from the original deposit [10]. Insurance terms and vulnerability parameters in the simulation are illustrative. Claims, repair invoices and insurance schedules were not used.

Appendix. Illustrative results from the legacy prototype

The following results reproduce the earlier experiment. Its latent headroom and measurement errors were specified in advance. The estimates illustrate how to report the paired comparisons and provide no evidence of an ADT advantage under the physical model specified above.

Matched-spending operating comparison

We simulate ten portfolios with the same exposure, vulnerability and contract in each paired comparison. At matched mean intervention spending, twin-guided operation reduces average annual covered loss by a median of 37% (book range 23–55%) and 99.5th-percentile annual covered loss by 13% (7–27%) (Fig. 3a). The comparison with a class rule using the same trigger margin yields a 58% average-loss reduction and a 25% tail reduction, with median intervention spending 70% higher for twin-guided operation. At matched spending, the median tail benefit across four paired books falls from 15% under the base calibration to 9% under the heavier-tail variant. In the model, each action adds a finite amount of headroom that severe heat can still exceed. The heavier-tail check shows how much the estimated benefit depends on the event generator.

Fixed-operation underwriting comparison

Holding the class-based operating rule fixed, we then change only the information used to estimate headroom in underwriting. The relative 99.5th-percentile loss error across the ten synthetic books has an RMSE of 30.2% with class evidence and 7.5% with twin evidence (Fig. 3b). The comparison is against the simulated loss for each book. With this operating rule, the median fraction of simulated years exceeding the modelled threshold changes from 0.18% with class evidence to 0.60% with twin evidence, against the nominal exceedance probability of 0.50%. A smaller estimation error can bring funding closer to its target while increasing exceedances in books for which the class estimate was conservative.

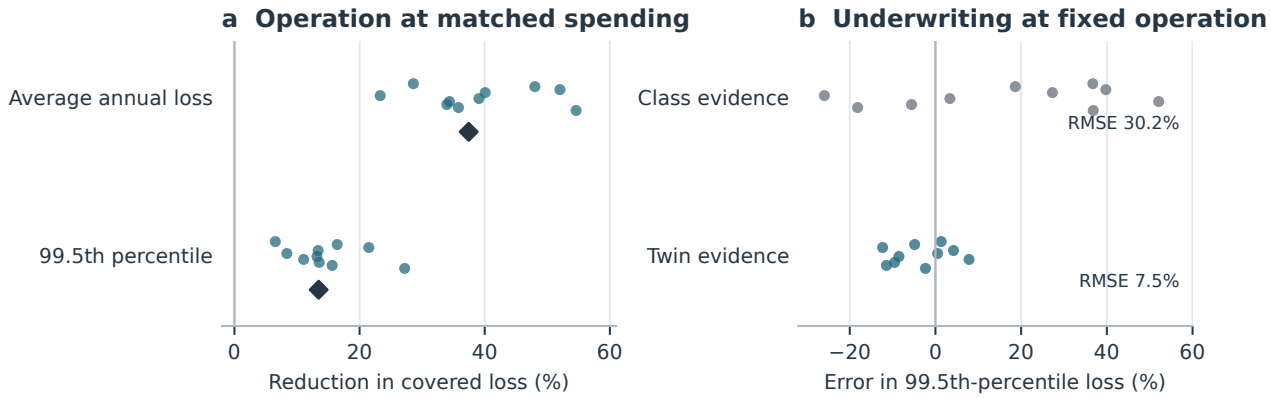
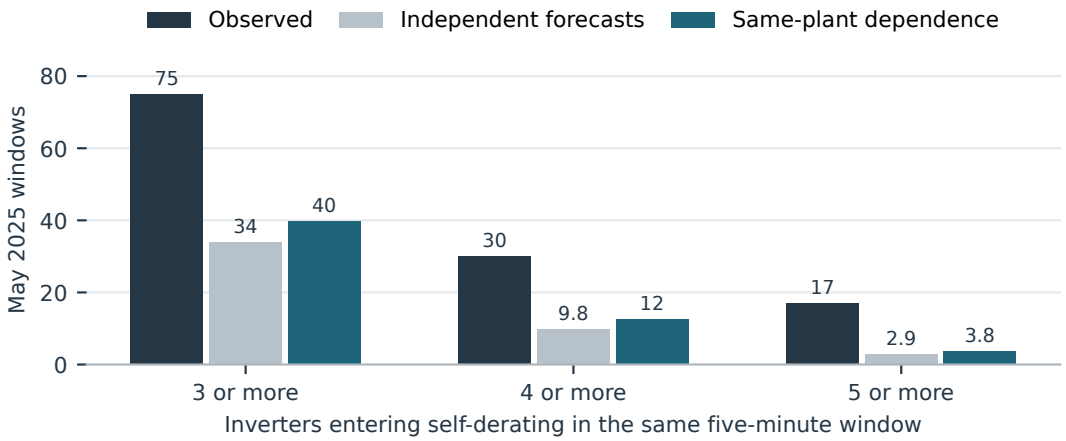


Fig. 3. Illustrative results from the assumed-headroom prototype. **a**, Reduction in covered loss with twin-guided operation relative to a class rule matched on average intervention spending. Dots represent synthetic books and diamonds mark medians. **b**, Relative error in annual 99.5th-percentile covered loss against simulated loss for the same book, holding class-based operation fixed. RMSE is across books. The panels use the base ERA5-calibrated event generator and the assumed breakdown contract in Table 2.



Extended Data Fig. 1. Coincident inverter self-derating in North Macedonia. Counts of May 2025 five-minute windows meeting each onset threshold. There were 75 windows with at least three starts, compared with 34 under conditional independence and 40 under a dependence model fitted to January–April records. The two forecasts use the same inverter-level probabilities.

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